

7.15 block 07b split receipt — 2026-09-26

Decision and provenance

The owner authorized three stacked replacement review units after all three Copilot rounds on firmware PR #863 returned findings. PR #863 remains open as the historical complete candidate. Its final code head is 729ae7cfacf581ce534a98f6f5e0cc39efee3838, based on 8c654bcd8e1c8936a86aaa625a4ab78900a250e9. The reviews were 5323189342 (four findings), 5323944293 (two) and 5324788734 (one). All seven threads received source dispositions and are resolved. No fourth review is requested on the unsplit PR.

The #863 audit report and PDF were frozen before this authorization. Their "split pending" statement is a historical status; this receipt records the subsequent owner decision and replacement stack.

Replacement drafts: firmware [#865](https://github.com/BitHighlander/keepkey-firmware/pull/865), [#866](https://github.com/BitHighlander/keepkey-firmware/pull/866) and [#867](https://github.com/BitHighlander/keepkey-firmware/pull/867), with companion Python [#114](https://github.com/BitHighlander/python-keepkey/pull/114), [#115](https://github.com/BitHighlander/python-keepkey/pull/115) and [#116](https://github.com/BitHighlander/python-keepkey/pull/116).

Unit	Boundary	Recorded stack reference	Python companion
A-C formatters and interpolated intent	7a base	5c0090e038769ed36d627d4dca09cbfcb1df91d3	f813519c7c1163783b0e89b5cbd70536831f5e49
D groups, one-array iteration, optional visibility	A-C	d5d015d9cf71f4656a09017ca7543f8ec1904b6d	8fc1c2bc038db8ee2c192b162c3308234117275e
E embedded calls and assembled audit	D	69478c750d741612bd3b53cec2474aabc7d0d6e4	67e891214086f7c7c6ebc7f75f9e3e9f27ec96a0

The E row names its immutable pre-receipt code pin; its final report-bearing head is recorded in firmware PR #867. A-C and D include the later wire-test expectation corrections. Their adjacent behavior and Python pins are unchanged.

The E firmware code matches #863's tracked code; the tracked tree differs in the Python gitlink and the split audit reports and receipts. The E companion's tracked tree matches #228's df376eec6ea7d705f7a70bdd31210f8cf929f97e; its new SHA gives it a reviewable diff against the D companion. Both equivalences were checked with git diff --exit-code before adding this receipt. Each unit adds only the source it compiles or dispatches; D and E source is absent from the A-C unit, and E source is absent from D.

File placement

Each cell identifies the adjacent branch diff that owns that file's edits. For a shared file, edits introduced in A-C, D and E are distinct changes; later edits to an earlier line belong to the later diff. The three adjacent diffs account for the entire original #863 tracked change set. The Python gitlink identifies three companion branches with the same boundaries.

Tracked path	A-C	D	E
deps/python-keepkey	✓	✓	✓
docs/release/audit-units/715-07b-formatters-audit-20260925.md			✓
docs/release/audit-units/715-07b-formatters-audit-20260925.pdf			✓
docs/security/HANDOFF-ERC7730-715-FORMATTERS.md	✓	✓	✓
docs/security/HANDOFF-ERC7730-PHASE-E.md			✓
include/keepkey/firmware/eip712_stream.h	✓		
include/keepkey/firmware/erc7730_abi_stream.h	✓		✓
include/keepkey/firmware/erc7730_capabilities.h	✓	✓	✓
include/keepkey/firmware/erc7730_catalog.h	✓	✓	✓
include/keepkey/firmware/erc7730_field.h	✓		✓
include/keepkey/firmware/erc7730_format.h	✓		

include/keepkey/firmware/erc7730_program.h	✓		
include/keepkey/firmware/erc7730_workflow.h	✓	✓	✓
include/keepkey/firmware/ethereum.h	✓		
lib/firmware/CMakeLists.txt	✓		
lib/firmware/eip712_stream.c	✓		
lib/firmware/erc7730_abi_stream.c	✓	✓	✓
lib/firmware/erc7730_capabilities.c	✓	✓	✓
lib/firmware/erc7730_catalog.c	✓	✓	✓
lib/firmware/erc7730_field.c	✓		✓
lib/firmware/erc7730_format.c	✓		
lib/firmware/erc7730_program.c	✓		
lib/firmware/erc7730_workflow.c	✓	✓	✓
lib/firmware/ethereum.c	✓		
lib/firmware/fsm.c			✓
lib/firmware/fsm_msg_ethereum.h	✓	✓	✓
scripts/emulator/test_stack07_regressions.py	✓	✓	✓
unittests/firmware/CMakeLists.txt	✓		
unittests/firmware/erc7730_abi_stream.cpp	✓		
unittests/firmware/erc7730_catalog.cpp	✓	✓	✓
unittests/firmware/erc7730_field.cpp	✓		✓
unittests/firmware/erc7730_format.cpp	✓		
unittests/firmware/erc7730_program.cpp	✓		
unittests/firmware/erc7730_workflow.cpp			✓

Finding placement and disposition

Finding group	Unit	Disposition
D1 numeric signer constants; D3 signer text and decimals; D4 native Wanchain state; D5 token table precedence; D6 intent titles; D7 line splitting; D8 control bytes and oversized raw values; D12 root signer path; D16 signer marks and raw unit	A-C	Preload refusal or exact on-screen disclosure, with native and wire controls.
D10 parallel array pairing; D18 fixed indices into dynamic arrays	D	Bounds failures remain fail closed and are documented with wire cases.
D2 embedded intent; D9 literal callee; D17 refused inner definition; inner @.value without amountPath; depth-two and call binding	E	Refuse at preload or show blind, with inner-call and binding controls.
D11 table, text and decimal limits	A-C	The first companion mirrors preload bounds.
D11 iteration path limits	D	The second companion mirrors one-array iteration.
D11 embedded-call limits	E	The third companion mirrors inner-call capabilities.
D14 field title	D	Iteration fields use a numbered device-owned title.
D15 pager continuation test handling	E	Runtime tests filter continuation pages without de-duplicating them.
Copilot round 1: scalar formatter repeated in an iteration	D	Value path must walk the active array; positive/negative firmware and mirror tests.
Copilot round 1: stale phase counts, E status and EX test count	E	Handoffs and release report corrected.

Copilot round 2: display frames overwrote path classes	D	Dedicated <code>display_frames</code> , with a class-preservation test.
Copilot round 2: replay string reader accepted NUL	A-C	Independent reader rejects NUL, with a negative control.
Copilot round 3: signer native alias hid the token address	A-C	Signer alias and address precede the amount; maximum and Wanchain boundaries tested.

The original 102-agent audit recorded 42 findings only as clusters D1–D19. The raw per-finding manifest was not retained. D13 and D19 were not named in the surviving handoff; this receipt cannot reconstruct or assign individual raw findings from those clusters. The 14 later re-audit defects are recorded in the #863 report as behaviors and controls, not as a raw manifest. These historical evidence gaps remain explicit; no finding is marked clean merely because its original row is unavailable. The seven Copilot comments are individually assigned above and resolved on #863.

Validation ledger

Unit	Local native evidence	Host/device compiler evidence	Other gates
A-C	<code>firmware-unit 649/649</code> in a clean run directory; catalog, formatter, raw-overflow and replay controls	Two focused compiler/validator cases pass	First split CI found five stale wire expectations; corrected on A-C and inherited by D. New exact-head CI pending.
D	<code>firmware-unit 652/652</code> ; iteration and display-frame controls pass	Seven focused compiler/validator cases pass	First full wire run found two stale iteration titles; corrected six expectations. New exact-head CI pending.
E	<code>firmware-unit 656/656</code> ; tree-equivalent to #863 code	Compiler/catalog 23 pass, 2 skip locally without the official registry	#863 exact-head CI 36263598278 passed 16 required jobs; split-head CI pending.

The canonical 7b pre-push script fails on inherited duplicate `nanopb` options, an absent report-gate test in the 7a base, and its own absent `.cppcheck-build` path. A corrected `cppcheck` invocation found no issues. The owner authorized the #863 push with this exception recorded. The split branches retain that same inherited preflight condition. Runs on all three split worktrees passed submodule cleanliness, unit-test registration and clang-format 20, then failed only those three inherited checks. Corrected `cppcheck` also completed without findings on A-C and D; E source is tree-equivalent to the prior corrected run on #863. Hosted CI and review results for replacement PRs belong to their exact heads and will be added to the PR descriptions.